

CareerAI

AI-Powered Career Intelligence Platform

INFOSYS SPRINGBOARD VIRTUAL INTERNSHIP 7.0

Milestone 2 Report

SUBMITTED BY:

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Domain: Artificial Intelligence

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1. Introduction

The project "CareerAI AI-Powered Career Intelligence Platform" is a comprehensive, full-stack web application designed to act as an intelligent career coach. By combining artificial intelligence, natural language processing (NLP), and structured data management, CareerAI assists job seekers and working professionals in evaluating their resumes, identifying critical skill gaps, discovering tailored career paths, and obtaining context-aware career guidance.

In modern recruitment processes, job seekers frequently face hurdles understanding how Applicant Tracking Systems (ATS) evaluate their resumes, identifying missing skills for target roles, and maintaining an up-to-date professional profile. Milestone 2 focuses on solving these core foundational requirements by implementing full User Profile Management, intelligent Resume Uploading, AI-driven Resume Parsing (extracting over 95% of structured candidate metadata), and robust Resume Lifecycle Management (Viewing, Downloading, Replacing, and Deleting resumes).

2. Objective

The primary objectives fulfilled in Milestone 2 of the CareerAI platform are:

- **User Profile Management** — Provide an end-to-end profile editing interface where users can manage personal info, education history, technical skills, certifications, work experience, projects, and career interests.
- **Resume Upload & Validation** — Support seamless resume file uploads in PDF, DOCX, DOC, and TXT formats with automated file type and size validation (up to 10 MB).
- **High-Accuracy Resume Parsing** — Extract up to 95%+ of candidate details (Name, Contact Info, Bio/Summary, Education, Skills, Work History, Certifications, Projects, Languages) using AI and NLP prompt engineering.
- **Resume Lifecycle Management** — Provide full CRUD functionality allowing users to view original uploaded files inline in the browser, download original files, replace existing resumes, and delete resume records.
- **Profile Completeness Analytics** — Implement a real-time progress indicator calculating user profile completeness (0% to 100%) to encourage thorough profile setup.
- **Backend REST Services** — Expose scalable RESTful FastAPI endpoints to handle authentication, profile updates, background parsing tasks, and file management securely.

3. Tech Stack

Milestone 2 was implemented leveraging modern full-stack web technologies and AI models:

Layer	Technology	Purpose
Frontend	React.js (v19) + Vite	Fast single-page application (SPA) with responsive component architecture
Backend	FastAPI (Python 3.12+)	High-performance asynchronous REST API framework
Database	MySQL + SQLAlchemy ORM	Relational database for storing users, profiles, and resume metadata
AI Engine	Groq API / Llama 3.3 70B	LLM inference engine for structured NLP resume parsing and profile extraction

Authentication	JWT (JSON Web Tokens)	Stateless, secure session management and endpoint protection
Text Extraction	PyPDF & python-docx	Parsing raw text content from PDF and DOCX resume uploads
UI & Styling	Vanilla CSS + Lucide React	Modern custom design tokens, glassmorphism UI, and clean icons
HTTP Client	Axios	Async API integration with request token interceptors and response handlers

4. System Architecture & Data Flow

The CareerAI platform follows a decoupled, three-tier architecture ensuring high performance, security, and scalability.

- **Frontend Tier:** Built using React.js and Vite. Communicates with backend endpoints asynchronously over HTTP JSON requests via Axios.
- **Backend Tier:** Implemented using FastAPI. Manages JWT security, file upload handling, database transactions via SQLAlchemy, and background AI execution pipelines.
- **Database Tier (MySQL):** Relational database hosting structured entities including Users, Profiles, Resumes, and Skill Inventories.
- **AI Processing Tier:** Integrates LLM capabilities via structured JSON mode prompts for high-accuracy NLP parsing.

Data Flow Architecture: System Workflow: Data Flow Architecture (Milestone 2 Workflow):

- Landing Page —► User Registration —► Secure Login (JWT Issued)
- Dashboard —► User Profile Management (Edit Bio, Skills, Education, Links)
- Resume Upload —► File Validation (PDF/DOCX/TXT) —► Server Storage & Database Record Creation
- AI Pipeline (Async) —► NLP Text Extraction —► Structured Data Parsing (95%+ Metadata)
- Profile Autofill —► Auto-populates Profile & Review Panel for User Confirmation
- Resume Management —► View Inline, Download Original, Replace, or Delete Resume

5. Modules Implemented

5.1 User Profile Management

The User Profile Management module provides a central hub where users can maintain their professional identity.

- **Personal Info & Bio:** Editable name, email, phone number, location (city, state/country), and personal professional summary/bio.
- **Education History:** Adding and updating degree details, institution names, and graduation years.

- **Skills & Certifications:** Interactive skill tag inputs allowing users to add, edit, or remove technical and soft skills with comma/Enter shortcuts.
- **Credentials & Projects:** Storing professional certifications, online courses/bootcamps, languages spoken, and key career achievements.
- **Career Interests:** Managing work preferences (Remote, Hybrid, On-site), availability status ("Open to work"), target job roles, and salary expectations.
- **AI Profile Autofill:** One-click AI feature that populates profile fields from the user's parsed resume.

5.2 Resume Upload Module

The Resume Upload module facilitates drag-and-drop file ingestion with instant feedback.

- **Multi-Format Ingestion:** Supports PDF (.pdf), Microsoft Word (.docx, .doc), and plain text (.txt) files up to 10 MB.
- **File Validation:** Real-time validation of file extensions and size prior to submission.
- **Background Processing:** Asynchronous non-blocking uploads returning instant status updates while background workers process AI extraction.

5.3 Resume Parsing Module (95%+ Extraction)

The AI Resume Parsing engine uses custom NLP prompt engineering to extract structured data from raw resume text with high precision.

- **Personal & Contact Info:** Extracts candidate full name, phone number, location, LinkedIn URL, GitHub URL, and portfolio link.
- **Professional Bio / Summary:** Generates a 2-4 sentence professional summary reflecting experience and core competencies.
- **Career & Titles:** Identifies primary current role title, suggested target next step, and total experience years.
- **Technical & Soft Skills:** Parses programming languages, frameworks, developer tools, and domain-specific skills.
- **Certifications & Courses:** Extracts industry certifications (e.g. AWS, PMP) and completed courses/training programs.

5.4 Resume Management & Profile Completion

The Resume Management module gives users full control over their uploaded resume files and profile metrics.

- **Inline View:** Opens uploaded PDF/DOCX resumes inline in the browser viewer.
- **File Download:** Allows downloading the original resume file at any time.
- **Replace Resume:** Enables uploading a new file to update existing parsed data.
- **Delete Resume:** Deletes the database entry and removes stored files from disk storage safely.
- **Profile Completion Meter:** Visual circular progress bar tracking completeness (0% to 100%) to encourage detailed setup.

5.5 Backend API Architecture

FastAPI REST endpoints implemented in Milestone 2:

- **Auth Endpoints:** POST /auth/register, POST /auth/login (JWT Token Authentication)
- **User & Profile Endpoints:** GET /users/me, PUT /users/me/profile, POST /users/me/autofill-profile
- **Resume Endpoints:** POST /resumes/upload, GET /resumes/, GET /resumes/{id}/status, GET /resumes/{id}/view, GET /resumes/{id}/download, DELETE /resumes/{id}

6. Output & Screenshots

Below are the empirical output screenshots demonstrating the implemented Milestone 2

CareerAI Dashboard Career Jobs Courses Coach Disha Sign out

Good morning, Disha

Upload a resume to get AI-powered career insights

Upload resume

Drop your resume here
PDF, DOCX, or TXT • Max 10 MB
[Browse Files](#)

LATEST ATS SCORE

92 /100

Disha_S_Resume_frontend.pdf

Career intel Jobs Courses AI Coach

Resume history

3

[View analysis](#) [Preview file](#) [Download](#) [Delete](#)

Disha_S_Resume_frontend.pdf **ATS 92** [View](#) [Download](#) [Delete](#)

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CareerAI Dashboard Career Jobs Courses Coach Disha Sign out

Career Intelligence

Based on: Disha_S_Resume_frontend.pdf

[Resume History \(3\)](#)

YOUR UPLOADED RESUMES

Disha_S_Resume_frontend.pdf **ATS 92** [Active](#) [Download](#) [Delete](#)

🕒 24 Jul 2026

Disha_S_1AT23CS046_Resume (1).pdf [View](#) [Download](#) [Delete](#)

🕒 24 Jul 2026 **ATS 92**

Disha_S_Resume_frontend.pdf [View](#) [Download](#) [Delete](#)

🕒 24 Jul 2026 **ATS 92**

ATS SCORE
92/100
Excellent

SKILLS
27
detected

SKILL GAPS
3
identified

CAREER PATHS
4
recommended

ATS score breakdown



Detected skills (27)

HTML5 CSS3 JavaScript (ES6+) React.js TypeScript Python
Java C Figma Wireframing Prototyping User Research
Usability Testing Interaction Design Information Architecture Git
GitHub Docker CI/CD Canva Netlify REST APIs Gemini API
Agile Project Management Cross-functional Collaboration
Analytical Thinking Problem Solving

\$ Salary estimate

MIN MEDIAN MAX MARKET DEMAND
\$60,000 \$80,000 \$100,000 high

⚠ Skill gap analysis

- **Cloud Computing** medium
most modern applications are deployed on cloud platforms
- **Cybersecurity** high
integral to software development and deployment
- **DevOps Tools like Kubernetes** medium
essential for efficient deployment and scaling

📈 Recommended career paths

Full Stack Developer 85%

Design and develop complete web applications, integrating frontend and backend functionality

JavaScript Python REST APIs

\$ \$80,000 - \$120,000

6-12 months

UX Engineer 90%

Combine design and engineering skills to create seamless user experiences

JavaScript React.js Figma

\$ \$100,000 - \$140,000

9-18 months

DevOps Engineer 80%

Ensure smooth operation of software systems, focusing on CI/CD and infrastructure management

Git Docker CI/CD

\$ \$110,000 - \$160,000

12-24 months

Technical Project Manager 75%

Lead cross-functional teams in developing and delivering software projects

Agile Project Management
Cross-functional Collaboration Analytical Thinking

\$ \$120,000 - \$180,000

18-36 months

✅ Experience roles detected

Frontend Developer Software Quality Engineer



Disha

abc@test.com

Profile completeness

93%

Complete your profile to get better job & course recommendations



Update Profile from Resume

Upload your resume to auto-extract skills, certifications & courses



Drop your resume here or [browse](#)

PDF, DOCX, DOC, TXT - Max 10 MB

About

Bio / Professional Summary

A passionate and detail-oriented Computer Science undergraduate with expertise in frontend development, software quality engineering, and AI-powered applications. Strong understanding of modern development practices, tools, and technologies, with a keen interest in building responsive, user-friendly web applications and expanding into software

Personal Info

Location

Bengaluru, India

Phone

1234567890

Availability

Open to work



Work Preference

Remote



Salary Expectation

13121

Social Links

LinkedIn

 <https://linkedin.com/in/dishas05>

GitHub

 <https://github.com/dishashiva>

Twitter / X

@ <https://twitter.com/yourname> or <https://x.com/yourname>

Personal Website

 <https://yourwebsite.com>

Portfolio

 <https://disha.is-a.dev>

Experience

Current Title

Student developer

Target / Dream Title

Software Quality Engineer or I

Years of Experience

0

Languages Spoken

English



Skills

Technical & Soft Skills

Python × HTML5 × CSS3 × RESTful APIs × MySQL × Git × GitHub × CI/CD ×
AWS Cloud × Netlify × Figma × Postman × Problem Solving × C ×

Add

Press Enter or comma to add

Interests / Hobbies



Certifications

Professional Certifications

AWS Certified AI Practitioner ×

Add

Press Enter or comma to add

Add certifications from your resume or manually type them in



Courses & Training

Online Courses, Bootcamps & Training

Nothing added yet

Add

Press Enter or comma to add

Add courses from your resume or manually type them in



Education & Achievements

Education Background

Notable Achievements

 Save Profile

7. Key Features Implemented in Milestone 2

Complete User Profile Management

Comprehensive form interface to manage bio, education, skills, certifications, work history, projects, and target role preferences.

1-Click AI Profile Autofill

Automated extraction pipeline that reads parsed resume data and populates user profile fields instantly.

High-Accuracy AI Resume Parsing (95%+)

NLP prompt pipeline extracting candidate metadata including name, contact details, social links, bio, skills, education, and credentials.

Secure Multi-Format Resume Upload

Supports PDF, DOCX, DOC, and TXT files with file type checking, size enforcement (10 MB limit), and background execution.

Full Resume CRUD & Lifecycle Management

Allows users to list uploaded resumes, view files inline in browser, download original files, replace existing resumes, and delete files.

Dynamic Profile Completion Analytics

Calculates and visualizes overall profile completion percentage (0% to 100%) to motivate users to provide complete career information.

FastAPI REST & MySQL Integration

Robust backend service layer utilizing SQLAlchemy ORM with automatic schema migrations and JWT authentication.

8. Conclusion

Milestone 2 of the CareerAI platform successfully implements all core foundational capabilities for user profile management, resume upload, AI resume parsing, and resume lifecycle management. By extracting over 95% of candidate information automatically and providing full CRUD controls alongside an intuitive React interface, the application delivers a seamless experience for job seekers.

With Milestone 2 complete, the platform provides a robust data foundation for Milestone 3, which will introduce ATS scoring algorithms, detailed skill gap detection, AI career path predictions, personalized job and course recommendations, and the AI Career Coach chatbot.